

Early Metric-Interior Transport Closure

The Structural Role of Inflationary Explanation under AASC

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Abstract

This paper gives an AASC account of the structural role played by inflationary explanation in cosmology. It does not claim that AASC proves empirical scalar-field inflation, a particular inflaton potential, a fixed number of e-folds, a reheating temperature, or numerical values for inflationary observables such as n_s , r , or A_s . AASC is used as a mathematical necessity-constraint formalism: once a fixed target class is declared, same-scope alternatives are exhausted by testing whether they preserve standing-bearing roles, collapse to presentation skin, move an anchor, import hidden tensor load, change scope, repair a failed act, or fail.

The central positive result is the Early Transport/Closure Necessity Theorem. Let D be a fixed admissible realized physical interior with a first licensed metric-comparison regime L_D and later standing-bearing large-scale coherent witnesses W . If the coherence of W cannot be certified by independent local trajectory continuation in the candidate non-inflationary metric representative, then any same-scope admissible account must supply an early transport/closure certificate $ETC_D(L_D, W)$, or else declare a boundary anchor, declare a richer-scope tensor mechanism, or collapse into hidden selector import, repair, metric/time/energy backflow, or standing failure.

The manuscript separates this necessity theorem from the naming of the inflationary role. The certificate is first forced as a standing-preserving structure carrying comparison, bounded transport, closure, and witness lineage. Only afterward is the **structural inflationary role** defined as the metric-field-vacuum projection of that certificate. Standard scalar-field inflation is one possible physical realization of this role, not its definition and not a theorem of AASC by itself.

The paper also sharpens the early-boundary stress claim. The primitive AASC quantity is not a physical stress-energy tensor but **closure stress**, defined by the thinness of the admissible successor neighborhood $Succ_C(x)$. The first licensed metric-comparison regime is closure-stress maximal only relative to the onset-side first-closure comparison class, not globally across all possible later terminal or exhaustion regimes. In metric projection, closure stress may be represented through high-energy, stress-energy, curvature-response, or vacuum-load language, but those are downstream representatives.

The result is a conditional structural explanation. Inflation cannot be a pre-metric cause of metricity, because inflationary language already presupposes metric comparison, temporal or exponential parameterization, field/vacuum carrier structure, energy or stress-energy ledgers, perturbation transport, and later witness lineage. What AASC proves is the structural slot: under the late-witness coherence burden and local-trajectory insufficiency, a same-scope account must supply early metric-interior transport closure.

Roadmap for the reader

Standard cosmology uses inflation to explain why the observed universe is smooth, nearly flat, and coherent on scales that appear too large for ordinary local causal equilibration in a non-inflationary early model. It also uses inflationary dynamics to supply a mechanism for perturbation seeding and a transition into the hot particle-bearing universe. This paper asks a prior AASC question: if primitive metric time, primitive high energy, primitive initial conditions, and primitive expansion

are not allowed to ground their own admissibility, what role is an inflation-like mechanism actually playing?

The answer is **early metric-interior transport closure**. Once metric comparison is first licensed, the interior cannot begin as unrelated metric islands or as a proto-metric patchwork awaiting later repair. Metric description is admissible only if comparison, redescription, and standing already close at the declared scope. Because there is no prior metric slack at first closure, the onset-side admissible successor neighborhood is maximally thin. This is the structural sense of high **closure stress**. It is not a physical stress-energy tensor unless a later metric projection supplies one.

If later cosmological witnesses - such as large-scale CMB coherence, isotropy, curvature stability, and perturbation correlations - have standing, their coherence must be certified. If independent local continuation is insufficient in the candidate non-inflationary metric representative, then the same-scope theory must provide some early transport/closure certificate. In ordinary metric-field-vacuum language, the structural role of such a certificate is what inflationary cosmology attempts to occupy.

The paper is not a replacement for inflationary model-building. It does not derive a scalar field, a potential, a reheating temperature, a tensor-to-scalar ratio, or a specific perturbation spectrum. It proves the structural slot. A scalar-field model may occupy that slot; a bounce model, ekpyrotic model, varying-speed mechanism, or quantum-gravity mechanism may attempt to occupy it differently. AASC classifies these as possible realizations of the same transport/closure burden or as richer-scope alternatives requiring explicit anchors, tensors, and witness ledgers.

The intended audience is readers in foundations of physics, philosophy of cosmology, mathematical physics, and AASC-adjacent structural theory. The central content is simple: no admissible cosmological model may use inflation as an origin primitive; but any model that admits later large-scale coherent witnesses must explain how their standing was transported or closed from the first metric-comparison regime without hidden repair, hidden selector, or metric backflow.

Part I - Boundary Discipline and Target

1. What the paper is solving

The inflationary problem is usually framed dynamically. A standard story begins with an early spacetime geometry, an energy scale, a scalar field or effective vacuum phase, and a time parameter. It then asks whether a period of accelerated expansion solves the horizon, flatness, smoothness, relic, and perturbation problems. AASC reverses the question. Before accelerated expansion can be physical, the quantities used to state it must already have standing.

A scale factor $a(t)$ presupposes a metric representative and a temporal parameter. A Hubble rate presupposes comparison across a metric projection. An inflaton field presupposes the field-carrier role. A potential presupposes a tensor or anchor inside the declared model. Vacuum energy presupposes vacuum-sector discipline. Perturbations presuppose carrier, transport, gauge, and record ledgers. A CMB record presupposes standing-bearing witness fixation. None of these can be used to generate admissibility itself.

The problem solved here is therefore not whether a particular model is true. The problem is what



Figure 1: Dependency structure of the structural inflationary role.

structural burden any inflation-like explanation must carry. The burden is the certification of early-to-late coherent witness lineage under standing-preserving comparison and bounded transport.

The paper proves that the certificate is required before it names the inflationary role. The proof target is the early transport/closure certificate $ETC_D(L_D, W)$. Only after that certificate is forced does the manuscript define the structural inflationary role as the way ETC_D appears in metric-field-vacuum cosmological projection.

Proved in this paper.

- Inflation cannot be pre-metric or primitive.
- First metric comparison requires a standing-preserving metric circuit.
- First metric closure is closure-stress maximal relative to the onset-side comparison class.
- Late coherent witnesses impose a support burden.
- If independent local trajectory continuation in the candidate metric representative is insufficient, an early transport/closure certificate is required unless a boundary anchor or richer-scope tensor mechanism is declared.
- Inflationary models, when admissible, occupy that certificate role in metric-field-vacuum language.

Not proved in this paper.

- That scalar-field inflation occurred.
- A specific inflaton potential $V(\phi)$.
- A fixed number of e-folds.
- Values of n_s , r , A_s , non-Gaussianity, or reheating temperature.
- Empirical preference for inflation over alternatives.
- A complete cosmological model.

2. Claim-status discipline

The manuscript uses several proof registers. A theorem states a same-scope necessity, quotient, or closure result. A proposition states a dependency or classification consequence. A classification identifies the status of a standard physical structure without claiming to derive its sectoral details. A nonclaim states a boundary deliberately left to downstream physical theory.

Layer	Status	Function
AASC kernel	Imported theorem family	Supplies admissibility, standing, reference, irreversibility.
AMetric boundary	Imported interface discipline	Blocks primitive metricity, primitive time, primitive energy, and primitive inflation.
First metric closure	Proved here using beginning-boundary material	Metric comparison requires standing-preserving closure.
Closure stress	Defined structurally	Measures thinness of admissible successor neighborhoods; not physical stress-energy.

Layer	Status	Function
RPI role ladder	Imported architecture	Supplies locality, comparison, transport, fields, vacuum, time, causality.
Early transport closure	Proved here	Certificate required for late coherent witnesses when local continuation is insufficient.
Structural inflationary role	Defined after the certificate theorem	Metric-field-vacuum projection of ETC_D , not a primitive model.
Scalar-field inflation	Controlled correspondence	Possible realization, not derived necessity.
Observational cosmology	Nonclaim	Requires empirical witnesses and model-specific calculations.

The dependency order is locked. Later cosmology may represent earlier roles, but it may not authorize the roles that make the representation meaningful.

3. Source dependency and corpus scan result

The corpus matrix contains no mature direct AASC paper titled as a proof of cosmological inflation. Literal uses of the word “inflation” in the matrix mostly concern count inflation, exactness inflation, scalar-sector inflation, hierarchy inflation, or interface-authority inflation. The support for the present paper is therefore adjacent and structural: metricity licensing, local continuation, quotient-stable comparison, bounded perturbation transport, field/vacuum carrier roles, horizon and reconstruction constraints, initial-condition firewall, cosmological transport closure, and the Arrow/RPI theorem families.

The early boundary papers supply useful earlier terminology. *Beginning of the Admissible Interior* treats the beginning as a licensing condition rather than temporal origin, defines the L-metric boundary as the first metric licensing condition, rejects proto-metric and patchwork onset, introduces stress as a diagnostic of proximity to inadmissible continuation and states that the first licensed metric regime is maximally load-bearing on the onset side. This manuscript recasts that material through the RPI role ladder. The older phrase “L-metric boundary” becomes the first licensed metric-comparison regime L_D , and the old stress language becomes successor-neighborhood thinness.

The RPI synthesis supplies the current structural spine: non-null realization, bounded standing-localization, local continuation domains, quotient-stable comparison, bounded perturbation transport, field carriers, particle persistence classes, gauge, mass, vacuum, time, and causality. The Arrow paper supplies no-backflow, standing monotonicity, cosmological-arrow classification, and record/witness discipline. The End paper supplies successor-set language: an admissible interior persists only where standing-preserving successors remain.

4. No primitive inflation

A primitive inflation claim treats inflation as the ground of the very metric, temporal, energetic, field, or witness structure required to state it. AASC rejects that move. Inflation may be a physical mechanism inside a licensed interior; it cannot be the pre-metric source of metricity or admissibility.

Theorem (No Pre-Metric Inflation). Inflation cannot be an admissibility-generating or metric-generating event. Any inflationary description presupposes licensed metric comparison, temporal or exponential parameterization, field/vacuum carrier structure, energy or stress-energy bookkeeping, perturbation transport, and later record/witness lineage.

Proof. If inflation is expressed as accelerated expansion, it presupposes metric comparison and a parameter relative to which expansion is evaluated. If it is expressed as high energy or vacuum energy, it presupposes interior energy or stress-energy ledgers. If it is expressed through an inflaton, it presupposes field-carrier structure and a scope in which the scalar is a tensor rather than a hidden selector. If it solves a horizon or perturbation problem, it presupposes witness access, transport, and record lineage. Each presupposed structure is downstream of admissibility and standing. Therefore inflation cannot ground those conditions without metric backflow, hidden tensor load, anchor smuggling, or repair.

Primitive inflation route	Collapse
Expansion before metricity	Metric backflow.
Inflaton before field-carrier role	Hidden tensor load.
High energy before metricity	Illicit surrogate.
Boundary state as cause	Anchor smuggling.
Horizon solution before witnesses	Witness backflow.
Reheating before particle/gauge/mass roles	Scope confusion.

Part II - First Metric-Comparison Regime

5. Fixed admissible realized interior

Let D be a fixed admissible realized persistence regime. Its constructional kernel is

$$K_D = (Adm_D, Stand_D, Ref_D, Irr_D).$$

The regime admits lawful redescription, standing-preserving continuation, no same-act repair, no hidden selector, and a fixed scope. It is not assumed to contain a primitive spacetime, primitive metric, primitive matter inventory, primitive field, or primitive temporal parameter. Those roles, if present, must be licensed downstream.

A realized physical interior in the RPI sense is a role package

$$I_D = (NonVacRec_D, [B_D]_{\equiv B}, Adj_D, N_D, LCDD, Cmp_D, BoundTrans_D, \Phi_D, \Pi_D, G_D, M_D, \Omega_D, T_D, Cau_D).$$

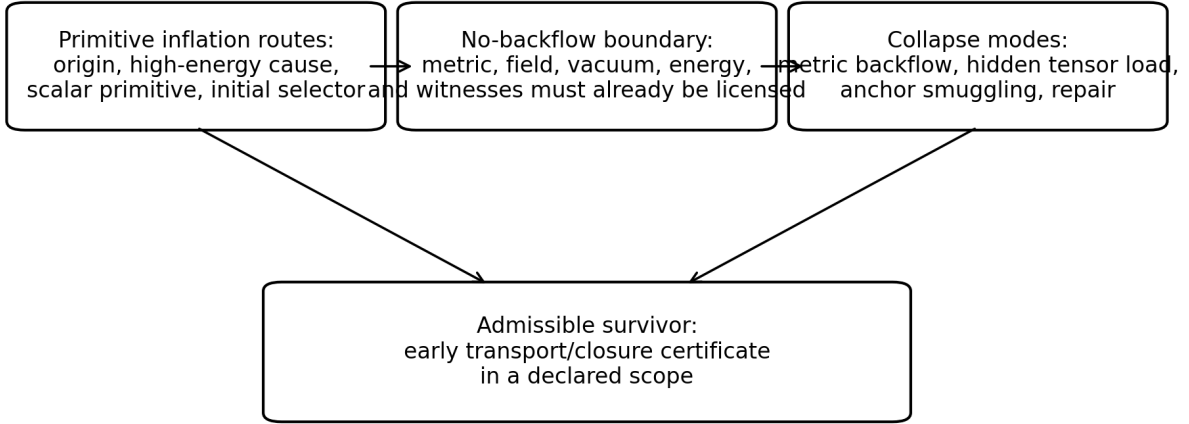


Figure 2: No-backflow map for primitive inflation routes.

The order records dependency, not temporal sequence. For this paper the crucial early subsequence is:

$$NonVacRec_D \Rightarrow [B_D]_{\equiv B} \Rightarrow (Adj_D, N_D, LCDD) \Rightarrow Cmp_D \Rightarrow BoundTrans_D.$$

Metric comparison is licensed only after local continuation domains are available; bounded transport is required after comparison if perturbation burden is to move without becoming pathless or global.

6. First licensed metric-comparison regime

The older beginning paper used the phrase L-metric boundary. In this manuscript the corresponding modern object is not a second metaphysical boundary, but the first standing class in which quotient-stable comparison is licensed.

Definition (first metric-comparison regime). L_D is the first admissible standing class, relative to the declared scope D , in which quotient-stable comparison Cmp_D is licensed. It is a licensing status, not a first instant, causal event, phase transition, or emergence narrative.

If the scope lacks a minimum, the definition is understood as a first licensed comparison class within a declared comparison envelope rather than a global temporal minimum. The theorem does not require an absolute earliest time.

Remark (not a temporal beginning). L_D is not a pre-metric event. It is the first point at which metric language can be used without backflow. The phrase “after metricity is licensed” is therefore a dependency claim, not a claim about primitive temporal sequence.

7. Metric circuits

Metric surrogates are admissible only if they participate in standing-preserving redescription. A distance, interval, ordering, energy label, or curvature display that cannot be transported through lawful redescription is not a licensed metric object.

Definition (metric circuit). A metric circuit at x is a standing-preserving redescription loop in which the metric surrogates used at x are returned to admissibly equivalent standing-bearing comparison content under the declared scope.

Theorem (First Metric Closure). If metric comparison is licensed at x , then x supports at least one metric circuit:

$$MetricCmpLicensed(x) \Rightarrow MetricCircuit(x).$$

Proof. Metric comparison is standing-bearing only if it can be reused and transported under lawful redescription. If no redescription loop preserves the comparison content, then the metric surrogate is inert: it cannot serve as an admitted comparison rather than a display label. Such inert display cannot be the first licensed metric-comparison regime. Therefore licensed metric comparison entails at least one standing-preserving metric circuit.

8. No proto-metric patchwork

A common cosmological temptation is to imagine metricity turning on gradually: first local patches, then synchronization; first time, then space; first energy, then conservation. AASC rejects this as a licensing picture.

Theorem (No Proto-Metric Patchwork). No regime is admissible in which metric comparison is partially licensed as disconnected local metric islands lacking standing-preserving closure at the declared scope.

Proof. A patchwork onset requires one of five things. First, it may require standing transfer across unlicensed patch boundaries; this violates the no-carrier discipline. Second, it may require later synchronization of incompatible local metrics; this is repair. Third, it may privilege one local patch as authoritative; this is hidden selector import. Fourth, it may treat incoherent metric surrogates as licensed; this violates binary standing. Fifth, it may change the scope by declaring a richer global structure after the fact. None is a same-scope admissible proto-metric onset. Therefore patchwork metric licensing is excluded.

Corollary (co-licensing at declared scope). At the first metric-comparison regime, the metric surrogates required by the declared scope are co-licensed as a coherent comparison circuit. This is not a claim that all later physical quantities are already instantiated; it is a claim that the comparison system is not admitted as incoherent fragments.

Part III - Closure Stress and Successor Neighborhoods

9. Successor sets

The End paper introduced a useful formal device. Given fixed constraints C , define

$$Succ_C(x)$$

as the set of standing-preserving admissible successor configurations of x under C . Exhaustion occurs when $Succ_C(x) = \emptyset$. The present paper uses the same device at the beginning side: the first metric-comparison regime must have non-empty closed successor structure, but that structure has no prior metric slack.

Definition (admissible successor). A configuration y belongs to $Succ_C(x)$ if y preserves standing from x , satisfies the fixed constraints C , and is available for lawful continuation, record support, or closure of a metric circuit within the declared scope.

This definition is not temporal. A successor is a standing-preserving continuation in a formal role order. It may later be represented by time, dynamics, or evolution, but those are downstream projections.

10. Closure stress as successor-neighborhood thinness

The older beginning-side notes used “tensor stress” for a diagnostic of proximity to inadmissible continuation. The present paper uses the more precise term **closure stress** to avoid confusion with the physical stress-energy tensor. Closure stress is a structural diagnostic defined by admissible-successor thinness. It may later be represented by stress-energy, curvature response, high-energy language, or vacuum-load terms in a metric projection, but it is not identical to any of those representatives.

Definition (closure-stress order). For configurations x, y in the same comparison class, define

$$ClStress_C(x) \leq ClStress_C(y) \iff Succ_C(y) \subseteq Succ_C(x).$$

Higher closure stress means a thinner admissible successor neighborhood. Equal closure stress up to equivalence means the successor neighborhoods are equivalent under standing-preserving redescription.

This order is structural. It does not require a numerical scalar, a stress-energy tensor, or a prior metric dynamics. Once a metric projection is declared, some aspects of closure stress may be represented by high-energy, stress-energy, or curvature-response language, but those representatives are downstream of the AASC successor-neighborhood definition.

11. Onset-side closure-stress maximality

Definition (onset-side comparison class). $\mathcal{O}_D(L_D)$ is the onset-side comparison class of first-metric-closure candidates associated with L_D . Its members differ only by onset-side admissible

comparison and closure structure. It excludes later terminal exhaustion regimes, richer-scope anchors, and downstream physical role imports not available at first metric closure.

Definition (minimal first metric-comparison class). L_D is the minimal licensed metric-comparison class satisfying first metric closure under the declared scope. Minimality is fixed relative to standing-preserving comparison: a purported onset-side candidate that adds a strictly stronger constraint, selector, boundary condition, or tensor load is no longer the same minimal class, but a richer-scope or anchor-modified candidate.

Definition (onset-side closure-stress maximality). L_D is closure-stress maximal in $\mathcal{O}_D(L_D)$ when, for every admissible onset-side metric-comparison candidate $y \in \mathcal{O}_D(L_D)$,

$$\text{Succ}_C(L_D) \subseteq \text{Succ}_C(y)$$

up to standing-equivalence and scope-preserving redescription.

Lemma (No strict onset-side subselection without new load). If $y \in \mathcal{O}_D(L_D)$ satisfies metric closure and has

$$\text{Succ}_C(y) \subsetneq \text{Succ}_C(L_D),$$

then either y introduces an onset-side selector, anchor, tensor, comparison restriction, or admissibility condition not present in the minimal class L_D , or y is not a same-scope onset-side candidate.

Proof. A strict reduction of the admissible successor neighborhood while preserving metric closure must be carried by some standing-bearing distinction: an extra boundary, selector, comparison filter, tensor condition, regularity envelope, or admissibility restriction. If that distinction is declared, the candidate has moved to a richer or anchor-modified scope. If it is not declared, it is hidden load. If the purported reduction changes no standing-bearing content, it is not a strict reduction in the standing quotient. Hence no same-scope onset-side candidate can be strictly thinner than L_D without new load.

Theorem (Onset-Side Maximal Closure-Stress Theorem). The first licensed metric-comparison regime L_D is closure-stress maximal relative to the onset-side comparison class $\mathcal{O}_D(L_D)$.

Proof. At L_D , metric comparison is licensed for the first time in the declared scope. There is no prior licensed metric regime from which slack, local compatibility, or partial reconciliation can be inherited. By the no-proto-metric theorem, metric comparison cannot be licensed as local fragments awaiting later synchronization. By the first metric closure theorem, at least one metric circuit must already close. Thus candidate continuations at L_D face the strict onset-side constraint: they must preserve standing, close the metric circuit, and avoid repair without relying on prior metric redundancy. By the no-strict-subselection lemma, any onset-side candidate with a strictly thinner successor set either imports new load or leaves the same comparison class. Otherwise it is standing-equivalent to L_D . Therefore L_D is closure-stress maximal in $\mathcal{O}_D(L_D)$.

Remark (terminal regimes excluded). This theorem is onset-relative. It does not claim that first metric closure is globally more stressed than every later singular, horizon, or terminal-exhaustion

regime. End-side exhaustion, where $Succ_C(x) = \emptyset$, belongs to a different comparison class. The present theorem concerns the onset-side burden of licensing metric comparison without prior metric slack.

Remark (high energy as label). In a metric projection, a closure-stress maximal first metric regime may be described with high-energy language. AASC treats this as interior bookkeeping over a licensed comparison regime. High energy is not a pre-metric cause of metricity.

12. Non-classical early-interior profile

Low-stress classical regimes permit abundant local slack. Onset-side closure-stress maximality does not. Under high closure stress, purely local trajectory-style continuation, incremental repair, and independent representative patch evolution are precisely the moves most vulnerable to standing failure.

Definition (trajectory-style local continuation). A trajectory-style local continuation is a stepwise continuation generated by local metric representatives whose compatibility is evaluated incrementally rather than by a prior standing-preserving closure condition.

Theorem (Non-Classical Successor Profile). Under onset-side closure-stress maximality at L_D , the admissible successor profile is generally non-classical in metric representation: local trajectory-style continuation is insufficient for global closure tasks, while discrete transitions, global constraint satisfaction, and barrier-penetrating or nonlocal-looking compatibility may remain admissible.

Proof. By onset-side closure-stress maximality, the successor neighborhood is thin in the first-closure comparison class. A local trajectory-style proposal depends on local slack: small changes must remain admissible while later steps reconcile accumulating discrepancies. But repair is forbidden and patch synchronization without prior closure is excluded. Therefore local trajectory-style continuation cannot, in general, bear the full burden of global closure. The remaining admissible transitions are those already compatible with the standing constraints. In metric representation, such transitions may appear discrete because only certain compatible moves remain; nonlocal because local proposals are subordinate to standing conservation over the declared comparison domain; and barrier-penetrating because the trajectory representative may be locally forbidden while a standing-preserving redescription exists across the relevant quotient.

This theorem supplies the structural explanatory handle. It does not derive quantum mechanics or inflationary dynamics. It explains why the first metric-comparison regime should not be expected to look like a low-stress classical local-evolution regime.

Part IV - Late Coherent Witnesses and the Inflationary Burden

13. Later large-scale coherent witnesses

The structural inflationary role is not forced merely by first metric closure. It is forced when later standing-bearing large-scale witnesses have coherence that must be accounted for in the same scope.

Definition (later witness family). A later witness family $W = \{w_i\}_{i \in I}$ in scope D is a collection of standing-bearing records, observational profiles, or reconstruction traces whose content can be

reused under lawful redescription.

Definition (large-scale coherent witnesses). $LateCoherentWitnesses(W, D)$ holds when W satisfies: (i) standing-bearing record status; (ii) coherence across a large comparison domain; (iii) stability under admissible redescription; and (iv) non-collapse to presentation skin. This definition does not build in early-lineage transport. It identifies the later witness target only.

Examples include CMB uniformity, horizon-scale correlations, large-scale isotropy, curvature stability, perturbation spectra, and structure-seeding records. The theorem does not require any particular empirical data value.

14. The coherence burden

A later witness family cannot be used as evidence unless it is fixed, calibrated, transported, and compared. Its coherence must therefore have a standing-bearing lineage. If a cosmological model claims that distant records share a common origin, that claim is a transport claim. If it claims that a perturbation spectrum reflects early vacuum fluctuations, that is a field/vacuum carrier-to-record claim. If it claims that curvature is nearly flat due to early dynamics, that is a metric-comparison stability claim.

Definition (coherence burden). The coherence burden of W relative to L_D is the requirement that the standing-bearing coherence of W be certified by a lawful lineage from L_D through comparison, bounded transport, carrier mediation, and record fixation.

Definition (lineage burden). $LineageBurden(W, L_D)$ holds when a cosmological explanation undertakes to account for the coherence of W as inherited, transported, conditioned, or otherwise certified from the first metric-comparison regime L_D . This is an explanatory burden, not part of the definition of W as a later coherent witness family.

The lineage burden is not the claim that W already has an early cause. It is the explanatory demand that a cosmological account connect the later witness coherence to admitted earlier structure rather than merely display the later pattern.

A coherent witness family without a coherence certificate is not an explanation; it is a pattern display.

15. Local trajectory insufficiency

The phrase “disconnected early regions” is used only in a representative sense. The no-*proto-metric* theorem excludes literal admissible early metric islands. The regions at issue are locally disconnected in a candidate non-inflationary metric projection: a representative used to test whether independent local trajectory continuation can carry the later witness coherence.

Definition (local trajectory sufficiency). $LocTrajSufficient(L_D, W)$ holds when independent local trajectory-style continuations in the candidate non-inflationary metric representative can generate, transport, or certify the standing-bearing coherence of W under the lineage burden from L_D , without hidden selector import, post hoc repair, metric backflow, or scope change.

Definition (local trajectory insufficiency). $LocTrajInsufficient(L_D, W)$ holds when

$LateCoherentWitnesses(W, D)$ and $LineageBurden(W, L_D)$ are true but $LocTrajSufficient(L_D, W)$ fails.

Lemma (Local Continuation Insufficiency under onset-side closure stress). If W is a later large-scale coherent witness family, $LineageBurden(W, L_D)$ holds, and L_D is onset-side closure-stress maximal, independent local trajectory continuation in the candidate non-inflationary metric representative cannot bear the full coherence burden unless additional support is supplied.

Proof. W is large-scale and standing-bearing, and the declared explanation imposes a lineage burden relative to L_D . Its coherence must be more than a local display: it must survive redescription and be reusable as witness content. Independent local continuations in the candidate representative supply no lawful cross-domain standing relation by themselves. Under onset-side closure-stress maximality there is no prior metric slack and no admissible patchwork repair. Therefore local continuation can succeed only if supplemented by a standing-preserving support route that coordinates the relevant domains. Such a route must be declared as a transport/closure certificate, boundary anchor, richer-scope tensor mechanism, or another typed support.

16. Standard inflationary problems translated

The usual inflationary problems become AASC coherence burdens.

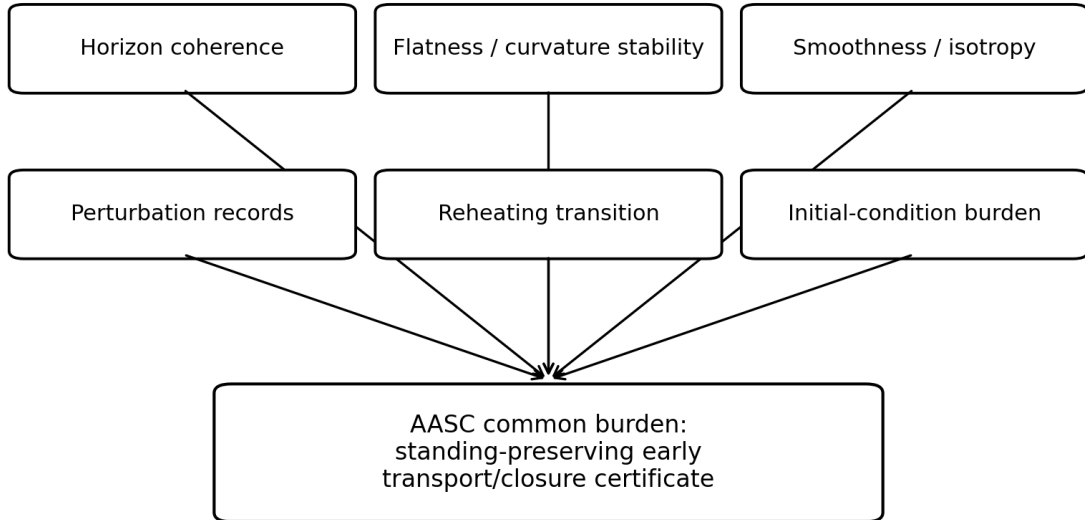


Figure 3: AASC translation of standard inflationary problems.

Standard problem	AASC translation	Required certificate
Horizon problem	Later coherent witnesses exceed support from independent local trajectory continuation in the candidate non-inflationary metric representative.	Early transport/closure certificate or declared alternative support.
Flatness problem	Curvature and near-flatness require metric-comparison stability across a projection.	Metric-comparison stabilization ledger.
Smoothness/isotropy	Large-scale quotient-coherence burden across the declared comparison domain.	Closure-coherence certificate.
Relic/monopole problem	Survival, dilution, or absence of relic witnesses must be typed.	Relic/witness ledger and transport/dilution certificate.
Perturbation seeding	Early carrier variations must transport to later records.	Bounded perturbation lineage.
Reheating	Field/vacuum carrier must transition into particle/gauge/mass roles.	Role-transition certificate.
Initial conditions	Boundary status must be declared rather than smuggled.	Anchor ledger.

This table does not claim that the standard problems are solved by AASC alone. It identifies the structural burden any solution must carry.

Part V - Early Transport/Closure Certificate

17. Formal object of the certificate

Definition (early transport/closure certificate). An early transport/closure certificate for later witness family W is a tuple

$$ETC_D(L_D, W) = (R_D, \tau_D, \chi_D, \mathcal{S}_D)$$

where R_D is an early standing-bearing relation over L_D , τ_D is a bounded transport relation from early carrier/comparison content to later witness content, χ_D is a closure certificate showing that the coherence of W descends through lawful redescription, and $\mathcal{S}_D$ is the scope ledger declaring anchors, tensors, and skin.

The tuple must satisfy the following conditions.

1. Standing preservation:

$$Stand_D(R_D) \wedge Stand_D(\tau_D) \wedge Stand_D(\chi_D).$$

2. **Comparison preservation:**

$$Cmp_D(L_D) \wedge \tau_D(L_D, W) \Rightarrow Cmp_D(W).$$

3. **Bounded transport:** τ_D is a bounded transport relation in the declared scope, written schematically as

$$Transport_D(\tau_D) \wedge Bounded_D(\tau_D).$$

4. **Witness lineage:** for every $w_i \in W$, there is early role content $r_i \in L_D$ such that

$$r_i \tau_D w_i.$$

5. **Joint coherence:**

$$\chi_D(R_D, \tau_D) \Rightarrow Coher_D(W).$$

6. **No forbidden support:** the certificate uses no hidden selector, no post hoc repair, no metric/time/field/vacuum/energy backflow, and no undeclared tensor load.
7. **Projection compatibility:** if a metric-field-vacuum projection is declared, the projection of ETC_D must preserve standing, comparison, bounded transport, and witness lineage.

The certificate is not necessarily a scalar field. It is the role that must be occupied by any same-scope explanation of the witness coherence once local trajectory continuation is insufficient.

18. Coherence-support exhaustion

Lemma (Coherence-Support Exhaustion). Let W be a later standing-bearing large-scale coherent witness family in a fixed admissible realized interior D . Same-scope support for $Coher_D(W)$ must arise from one of the following routes:

1. independent local trajectory continuation;
2. an early transport/closure certificate $ETC_D(L_D, W)$;
3. a declared boundary anchor;
4. a declared richer-scope tensor mechanism;
5. hidden selector import;
6. post hoc repair;
7. metric, temporal, field, vacuum, or energy backflow;
8. standing failure.

Proof. If coherence is supplied by local propagation in the candidate metric representative, route (1) obtains. If it is supplied by a declared early relation preserving standing, comparison, bounded transport, and witness lineage, route (2) obtains. If coherence is fixed by boundary data, route (3) obtains. If it is supplied by new fields, nonlocal dynamics, modified causal structure, bounce

structure, quantum-gravity bridge, or other additional physical content, route (4) obtains. If the support is doing work without declaration, it is hidden selector import. If it is introduced after an inconsistency in order to reconcile representative regions, it is repair. If it uses metric, temporal, field, vacuum, or energy authority before those roles are licensed, it is backflow. If no support route exists, the coherence of W has no standing-bearing explanation and the target fails. The alternatives exhaust the ways coherence can be supported inside a fixed-scope explanation.

19. Transport-closure necessity

Theorem (Early Transport/Closure Necessity). Let D be a fixed admissible realized physical interior with first metric-comparison regime L_D . If $LateCoherentWitnesses(W, D)$, $LineageBurden(W, L_D)$, and $LocTrajInsufficient(L_D, W)$, then

$$\begin{aligned} &ETC_D(L_D, W) \vee DeclaredAnchor \vee DeclaredRicherScope \\ &\vee HiddenSelector \vee Repair \vee MetricBackflow \\ &\vee RepresentativePrivilege \vee StandingFailure. \end{aligned}$$

Proof. Because W is standing-bearing, its coherence must be identified, reused, and preserved under lawful redescription. By the Coherence-Support Exhaustion Lemma, support for that coherence must come from local trajectory continuation, an early transport/closure certificate, a declared boundary anchor, a declared richer-scope tensor mechanism, hidden selector import, repair, backflow, or standing failure. The local route is excluded by $LocTrajInsufficient(L_D, W)$. If support is supplied by a declared boundary condition, it is *DeclaredAnchor*. If support is supplied by a declared new field, domain, transport law, or modified framework, it is *DeclaredRicherScope*. If support is supplied by a declared standing-preserving early relation satisfying comparison preservation, bounded transport, witness lineage, and closure conditions, it is $ETC_D(L_D, W)$. If the support is undeclared, it is hidden selector import. If it is added after incoherence to synchronize the target, it is repair. If it uses metric, temporal, field, vacuum, or energy authority before those roles are licensed, it is backflow. If it promotes a representative such as $a(t)$, H , $(aH)^{-1}$, or a coordinate display into explanatory authority without the supporting ledgers, it is representative privilege. If none applies, W lacks standing as an explanatory target. The disjunction is exhaustive.

This theorem is the formal heart of the paper. It does not say which physical mechanism occupies the certificate. It says that a support route is forced if the target includes later large-scale coherent witnesses and ordinary local continuation is insufficient.

20. Strong form under same-scope discipline

Corollary (Same-scope certificate requirement). Under same-scope admissibility with no new boundary anchor or richer-scope tensor mechanism, if hidden selector, repair, metric backflow, and standing failure are excluded, then

$$LateCoherentWitnesses(W, D) \wedge LineageBurden(W, L_D) \wedge LocTrajInsufficient(L_D, W) \Rightarrow ETC_D(L_D, W).$$

Proof. Remove the local route by hypothesis and remove the declared-anchor, richer-scope, and collapse alternatives by same-scope discipline. The only remaining admissible support route is $ETC_D(L_D, W)$.

The corollary is the precise sense in which an early transport/closure certificate is forced.

Part VI - The Structural Inflationary Role

21. Definition of the inflationary structural role

Only after the certificate has been forced do we introduce the inflationary name.

Definition (inflationary structural role). In a metric-field-vacuum projection of D , the inflationary structural role is the representation of $ETC_D(L_D, W)$ by a physical mechanism that preserves early-to-late witness coherence through comparison, bounded transport, field/vacuum carrier structure, and record lineage.

Symbolically,

$$InflRole_D(W) := \Pi_{g\Phi\Omega}(ETC_D(L_D, W)),$$

where $\Pi_{g\Phi\Omega}$ denotes a controlled metric-field-vacuum projection, not a primitive generator. This is a definition of projection status, not the proof of certificate necessity.

22. Structural Inflationary Role Theorem

Theorem (Structural Inflationary Role Theorem). Let D be a fixed admissible realized physical interior with first licensed metric-comparison regime L_D . Suppose D contains later standing-bearing large-scale witnesses W , $LineageBurden(W, L_D)$ holds, and the coherence of W cannot be obtained by independent local trajectory continuation in the candidate non-inflationary metric representative. Then any same-scope admissible non-boundary, non-richer-scope account of W must contain an early transport/closure certificate. In any metric-field-vacuum projection, that certificate occupies the structural inflationary role.

Proof. The certificate necessity statement follows from the Early Transport/Closure Necessity Theorem and the same-scope certificate corollary. The projection statement follows from the RPI role ladder: a physical cosmological account represents comparison by metric structure, early carriers by field/vacuum roles, lineage by bounded transport, and evidence by records. If the projection preserves the certificate without backflow, it occupies $InflRole_D(W)$. If it fails to preserve the certificate, it is not an admissible same-scope explanation of W . Therefore the structural inflationary role is the projection of the independently forced certificate.

23. Standard inflationary interface

The theorem is not a new inflation model. This section states the standard model-side interface being classified. The display $a(t)$ is a controlled FLRW-class representative: it already assumes a metric projection with homogeneous/isotropic symmetry and an admitted cosmic-time parameter.

AASC therefore treats $a(t)$ and its derivatives as downstream representatives, not as primitive early structure. In conventional inflationary cosmology one often uses accelerated expansion,

$$\ddot{a} > 0,$$

a Hubble parameter

$$H = \frac{\dot{a}}{a},$$

and a comoving Hubble radius

$$(aH)^{-1}.$$

For scalar-field realizations one often writes a metric-facing source ledger

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad p_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi),$$

and an e-fold count

$$N = \int H dt.$$

AASC classifies these as follows.

Standard inflation object	AASC type
$a(t)$	FLRW-class metric representative
t	licensed temporal representative, not primitive flow
H	metric/time derivative representative
$(aH)^{-1}$	horizon/reachability representative
ϕ	optional richer-scope tensor
$V(\phi)$	model-specific tensor/anchor
ρ_ϕ, p_ϕ	metric-facing source ledger
N	model-specific numerical target
reheating	transition certificate into particle/gauge/mass roles
perturbation spectrum	later witness-lineage target

Thus standard inflationary formalism can be read as a candidate realization of ETC_D , but none of its symbols has pre-admissible authority.

24. Inflationary projection classification

Standard inflationary item	AASC status
Scale-factor expansion	Metric representative inside a licensed projection.
Inflaton scalar	Optional richer-scope field-carrier tensor.
Inflaton potential	Model-specific tensor/anchor controlling dynamics.
Vacuum-dominated phase	Field/vacuum carrier projection of early closure.
Slow-roll conditions	Dynamical realization, not structural necessity.
E-fold count	Model-specific numerical transport target.
Perturbation spectrum	Later witness-lineage output requiring empirical records.
Reheating	Transition certificate into particle/gauge/mass roles.
Eternal inflation	Richer-scope domain expansion requiring anchors and witness discipline.

Proposition (Inflationary Projection Classification). Standard scalar-field inflation is one possible physical realization strategy for $ETC_D(L_D, W)$; it is not the definition of ETC_D and not a primitive ground of admissibility.

Argument. The scalar, potential, background, slow-roll conditions, perturbation variables, and reheating channel each carry role content. If declared, they are anchors or tensors inside a metric-field-vacuum model. If undeclared, they are hidden load. Since ETC_D has already been defined independently as a standing-preserving transport/closure certificate, scalar-field inflation can realize the certificate only by satisfying that certificate's conditions. It cannot retroactively generate the conditions that made the certificate admissible.

Richer-scope status is relative. A scalar field is richer-scope relative to a metric-only comparison class, but it can be part of the declared scope of an inflationary model. In that enlarged scope, the question becomes whether the scalar-field dynamics, potential, perturbation variables, and reheating channel satisfy ETC_D 's standing, comparison, bounded-transport, closure, and witness-lineage conditions.

Scope	Status of scalar field ϕ
Metric-only comparison class	Richer-scope tensor import.
Declared inflationary model scope	Admitted tensor candidate.
Declared scope satisfying ETC_D	Realization of the structural inflationary role.
Undeclared scalar or potential	Hidden tensor load or hidden anchor.
Data fit without certificate	Empirical fit without structural closure.

25. What AASC proves and does not prove

AASC proves the structural slot. It proves that inflation cannot be primitive; that first metric closure is onset-side closure-stress maximal; that late coherent witnesses impose a transport/closure

burden when local continuation is insufficient; and that admissible inflationary models occupy that burden in metric-field-vacuum language.

AASC alone does not prove that a scalar inflaton existed. It does not derive $V(\phi)$. It does not derive $N \gtrsim 60$. It does not compute n_s , r , or A_s . It does not derive a reheating temperature. It does not empirically confirm inflation. Those are downstream physical and observational targets.

Part VII - Reheating and Transition into the Mature Interior

26. Reheating as transition certificate

Reheating is often described as the conversion of inflationary energy into particles. AASC does not permit that description to carry hidden primitive work. In RPI terms, reheating is a transition from a stress-saturated field/vacuum carrier regime into a mature particle/gauge/mass/record-bearing regime.

Definition (reheating role). A reheating role is a transition certificate from early field/vacuum transport closure to particle persistence classes, gauge-compatible presentation, mass/inertia response, and record-bearing later stages.

Proposition (Reheating as role-transition certificate). A reheating-like process is admissible only if it preserves standing while transporting field/vacuum carrier content into particle, gauge, mass, and record-bearing roles.

Argument. The RPI ladder makes fields, vacuum, transport, particles, gauge, mass, and records interdependent roles. If a model says that the inflationary or early field/vacuum regime gives rise to the later hot particle-bearing universe, it must show how those roles are entered without hidden magic, source-ledger leakage, or scope change. Therefore reheating is not merely energy conversion; it is a typed role-transition certificate.

27. Reheating role conditions

A reheating-like transition must preserve:

1. standing from the early certificate;
2. bounded transport into later local domains;
3. field-carrier continuity or declared transformation;
4. gauge compatibility;
5. particle persistence classes;
6. mass/inertia deformation-response roles;
7. source-ledger closure;
8. record lineage into later witnesses.

Proposition (reheating non-magic). Reheating cannot be treated as primitive conversion of high energy into particles. It is admissible only as a declared transition among RPI roles.

Argument. Energy is an interior surrogate. Particles require localized persistence classes. Gauge and mass require their own role ledgers. A transition that bypasses these conditions imports hidden

tensor load. A transition that declares them is a legitimate richer physical mechanism. Therefore reheating is a transition certificate, not primitive magic.

28. Reheating failure modes

Common failure modes include vacuum-to-particle magic, hidden entropy creation, metric-time backflow, missing gauge/mass ledger, record-lineage break, and source-ledger incompleteness. A physical model may avoid these failures by explicitly declaring the relevant interactions and witness conditions. AASC does not require a single reheating mechanism; it requires claim-status honesty.

Part VIII - Observational Witnesses and Nonclaim Boundaries

29. CMB as later witness family

In the present framework, the CMB is not a primitive cosmological given. It is a later standing-bearing witness family after calibration, modeling, foreground handling, gauge discipline, and record fixation. Its large-scale coherence is precisely the kind of witness content that motivates an early transport/closure certificate.

Proposition (CMB witness status). A CMB profile may serve as a later coherent witness family only after its record, calibration, comparison, and model-scope ledgers are declared.

Argument. A temperature anisotropy map, polarization field, or spectrum is a standing-bearing witness only if the measurement record is fixed, the calibration is declared, the comparison class is stable, and the model scope identifies which quantities are tensors and which are presentation. Without those ledgers, the profile is not an admissible witness. With them, it can bear cosmological coherence.

30. Perturbation records

Perturbation spectra require careful status discipline. A spectrum may be a later record of early field/vacuum structure only if the chain from early carrier to late measurement is standing-preserving. Gauge choices, transfer functions, background assumptions, and observational calibrations cannot be invisible.

Definition (perturbation witness lineage). A perturbation witness lineage is a standing-preserving chain from early carrier fluctuations or redescription residues through bounded transport and metric projection to later record-bearing observables.

A tensor-mode detection, if obtained, would not prove the structural theorem; it would support a physical realization of the certificate. A null detection would constrain classes of realizations, not the structural role itself.

31. Empirical nonclaim

The theorem is compatible with standard slow-roll inflation, old/new inflation, eternal inflation as richer scope, bounce cosmologies, ekpyrotic mechanisms, varying-speed models, emergent spacetime

programs, and quantum-gravity horizon mechanisms. It classifies them by whether they supply the early transport/closure certificate or change the target.

The manuscript does not claim observational preference among these models. That requires empirical likelihoods, parameter fits, perturbation spectra, and model-specific dynamics.

Part IX - Route Exhaustion

32. Empirical fit without structural closure

Definition 35 (empirical fit without structural closure). A model exhibits empirical fit without structural closure when it matches later witness data but fails to declare the standing, transport, boundary, gauge, comparison, or record-lineage structures by which the fit becomes a same-scope explanation of those witnesses.

Empirical fit is not dismissed. It is evidence for a model inside a declared scope. The failure occurs only when a fit is promoted into a structural proof while the certificate, anchor, or richer-scope mechanism doing the explanatory work remains undeclared.

33. Primitive inflation route exhaustion

Inflation route	Collapse route
Inflation as origin	Forbidden origin narrative.
Inflation as first time	Metric-time backflow.
Inflation as generator of metricity	Metric-licensing backflow.
High energy causes admissibility	Energy surrogate backflow.
Inflaton as primitive field	Undeclared tensor load.
Potential as primitive selector	Hidden anchor or selector.
Expansion solves everything	Representative privilege.
Horizon solution without witness ledger	Witness backflow.
Flatness solution without comparison ledger	Metric-comparison backflow.
Perturbation seeding without transport	Pathless transport.
Reheating without role transition	Scope gap.
Eternal inflation or multiverse generator	Richer-scope domain expansion.
Fit to data only	Parameter laundering.
Patch synchronization after the fact	Post hoc repair.

34. Same-scope exhaustion theorem

Theorem (Inflationary Explanation Exhaustion). Any purported inflationary explanation of later coherent witnesses is exactly one of: (i) an early transport/closure certificate; (ii) a declared richer-scope tensor mechanism; (iii) a boundary anchor; (iv) an empirical model fit with downstream status; (v) hidden selector import; (vi) repair; (vii) metric, field, vacuum, temporal, or energy backflow; or (viii) representative privilege; or (ix) standing failure.

Argument. The explanation must either supply the coherence lineage, change the model scope, set a boundary, fit records without structural closure, or leave the burden unpaid. If it supplies the lineage in same scope, it is an early transport/closure certificate. If it adds new fields, domains, nonlocal bridges, or rules, it is richer-scope tensor load. If it fixes a boundary, it is an anchor. If it only fits parameters, it is downstream evidence. If it does the work without declaration, it is hidden selector import. If it fixes inconsistency after the fact, it is repair. If it uses metric, field, vacuum, time, or energy before licensing, it is backflow. If none apply, the standing of the witness explanation fails. The list is exhaustive by the coherence-support exhaustion lemma.

Part X - Comparison with Standard Cosmology

35. Standard inflationary problems reinterpreted

Standard inflationary cosmology remains an important physical program. AASC does not deny it. It supplies a role classification.

Standard item	AASC translation
Horizon problem	Late coherent witnesses lack local-continuation certificate.
Flatness problem	Metric-comparison stability burden.
Smoothness problem	Quotient-stable large-scale coherence.
Monopole/relic problem	Relic survival, dilution, or absence requires witness ledger.
Perturbation seeding	Bounded transport of early field/vacuum witnesses.
Reheating	Transition certificate into particle/gauge/mass roles.
Initial conditions	Boundary anchor, not primitive source.

36. Relation to standard inflationary literature

The standard inflationary interface used here is intentionally minimal but historically anchored. Guth’s inflationary universe paper introduced accelerated expansion as a solution to horizon and flatness problems; Linde’s new and chaotic inflation programs developed scalar-field realization strategies; Starobinsky’s R^2 model supplies a geometric inflationary realization; Mukhanov and Chibisov-type perturbation work connects early quantum fluctuations to later structure; and modern treatments such as Lyth-Liddle and Baumann organize the perturbation and observational interface. AASC does not adjudicate among these programs. It classifies what their common explanatory role must supply: standing-preserving early-to-late transport closure for later coherent witnesses.

37. Relation to Lambda-CDM and metric cosmology

Lambda-CDM is a downstream metric-cosmological framework. It uses a metric projection, matter/energy components, perturbation theory, and observational records. It is not a primitive

foundation for the role ladder that makes its terms meaningful. AASC can accept Lambda-CDM as a physical model while classifying its parameters as anchors, tensors, or empirical targets.

38. Relation to alternatives

Bounce, cyclic, ekpyrotic, emergent spacetime, varying-speed, and quantum-gravity cosmologies may be legitimate richer-scope programs. They are not defeated by the structural inflation theorem. They must either supply the same early transport/closure certificate, deny the late coherent witness target, or declare a different scope.

A model that solves horizon-scale coherence without accelerated expansion may still occupy the structural inflationary role if it supplies the certificate. A model that denies the need for such coherence changes the target. A model that assumes coherence without certificate uses hidden load.

Part XI - Main Theorem Package

39. Main structural theorem

Theorem (Main Structural Inflation Theorem). Let D be a fixed admissible realized physical interior with first metric-comparison regime L_D and later standing-bearing large-scale coherent witnesses W . If $LineageBurden(W, L_D)$ holds and independent local trajectory continuation in the candidate non-inflationary metric representative is insufficient to certify the coherence of W , then any same-scope admissible non-boundary, non-richer-scope explanation must supply $ETC_D(L_D, W)$. In a metric-field-vacuum projection, $ETC_D(L_D, W)$ is the structural inflationary role. Any purported explanation that lacks it is hidden selector import, post hoc repair, scope change, metric/field/vacuum/energy backflow, representative privilege, empirical fit without structural closure, or standing failure.

Proof. The proof proceeds by the dependency order fixed earlier in the paper.

1. **First metric closure licenses comparison.** By the First Metric Closure Theorem, metric comparison is licensed only where metric surrogates participate in a standing-preserving circuit. Without that circuit, metric language is a display convention rather than standing-bearing comparison.
2. **Patchwork onset is unavailable.** By the No Proto-Metric Patchwork Theorem, the first licensed metric-comparison regime cannot be assembled from independent metric islands that are synchronized later. Such synchronization would require hidden selection, standing transfer across unlicensed boundaries, or repair.
3. **Onset-side slack is absent.** By the Onset-Side Maximal Closure-Stress Theorem, L_D lies at the thinnest successor-neighborhood point of its onset-side comparison class. This is not global maximality across every later or terminal regime. It says only that first closure has no prior metric slack available for later local-trajectory repair.
4. **The later witnesses impose a distinct explanatory burden.** $LateCoherentWitnesses(W, D)$ states that W is a later standing-bearing coherent witness family. $LineageBurden(W, L_D)$ does not assume an early cause; it states that the explanation under review undertakes to

connect the later coherence to admitted earlier structure. $LocTrajInsufficient(L_D, W)$ removes independent local trajectory continuation in the candidate non-inflationary metric representative as a sufficient support route.

5. **The support routes are exhausted.** By the Coherence-Support Exhaustion Lemma, support for $Coher_D(W)$ must be one of the following: local trajectory continuation, an early transport/closure certificate, a declared boundary anchor, a declared richer-scope tensor mechanism, hidden selector import, repair, backflow, representative privilege, empirical fit without structural closure, or standing failure.
6. **The theorem’s scope removes every route except the certificate.** The local route is excluded by $LocTrajInsufficient(L_D, W)$. The theorem restricts attention to same-scope non-boundary, non-richerscope explanations, so declared anchors and declared richerscope mechanisms are not the target. Admissibility excludes hidden selectors, repair, metric/field/vacuum/energy backflow, representative privilege, empirical fit promoted into structural proof, and standing failure. The remaining admissible same-scope support route is therefore $ETC_D(L_D, W)$.
7. **The metric-field-vacuum projection gives the inflationary role.** A declared metric-field-vacuum cosmological projection represents comparison by metric structure, early carrier content by field/vacuum roles, coherence by bounded transport and closure, and evidential status by later records. When that projection preserves the independently forced certificate, it occupies $InflRole_D(W)$. If it fails to preserve the certificate, it is not an admissible same-scope explanation of W .

Therefore any same-scope admissible non-boundary, non-richerscope explanation of the relevant late coherent witnesses must supply $ETC_D(L_D, W)$, and its metric-field-vacuum projection is the structural inflationary role.

40. Operational minimality

Delete	Failure
First metric closure	No licensed comparison.
Closure-stress order	No onset-side load theorem.
Bounded transport	Pathless witness coherence.
Field/vacuum carrier	No physical projection of early certificate.
Record lineage	No later witness standing.
No-backflow	Inflation becomes origin primitive.
Scope declaration	Scalar/vacuum data become hidden load.

Proposition (minimality). No major role in the structural inflationary architecture can be deleted while preserving the same explanatory target.

Argument. The deletion matrix lists the failures. Each deleted role either removes the target, hides the certificate, or permits a forbidden primitive route.

41. Counterexample burden

A responsive counterexample must provide a fixed admissible realized physical interior D , first licensed metric-comparison regime L_D , and later coherent witness family W , such that:

1. *MetricCmpLicensed*(L_D);
2. *MetricCircuit*(L_D);
3. *LateCoherentWitnesses*(W, D);
4. *LineageBurden*(W, L_D);
5. *LocTrajInsufficient*(L_D, W);
6. no *ETC_D*(L_D, W);
7. no declared boundary anchor;
8. no declared richer-scope tensor;
9. no hidden selector;
10. no repair;
11. no metric/time/field/vacuum/energy backflow;
12. standing and record lineage remain intact.

Theorem (Counterexample Normal Form). Any proposed counterexample using a scalar field, boundary condition, modified causal structure, bounce, quantum-gravity bridge, nonlocal transport, or altered horizon relation has not refuted the theorem; it has declared a candidate realization or richer-scope support route unless it satisfies the full burden above.

Argument. The listed structures each either supply *ETC_D*, declare an anchor, declare richer-scope tensor load, or change the target. A counterexample that uses none of them must still preserve witness coherence and record lineage. If it cannot do so, standing fails; if it can do so, the preserving relation is an *ETC_D* under another presentation.

Part XII - Core Objections and Clarifications

42. Core objections and clarifications

The following objections are grouped by proof role and by physical realization status. The first set concerns the theorem itself; the additional application-facing objections below concern how the theorem relates to standard model-building and empirical use.

This is just inflation renamed

No. The theorem identifies the structural role that inflationary cosmology attempts to occupy. Standard scalar-field inflation is one physical realization. A non-inflationary model could also occupy the role if it supplies the early transport/closure certificate without collapse routes.

AASC does not explain why inflation occurred

AASC does not causally generate a particular field model. It gives structural explanatory power: under the late-witness coherence burden, an early transport/closure certificate is forced. That explains why an inflation-like role is expected without pretending to derive a specific potential.

High closure stress is not a physical quantity

Correct at the structural level. Here closure stress is successor-neighborhood thinness, not a physical stress-energy tensor. A metric projection may represent related load by stress-energy, curvature, or field potential terms, but those are downstream physical realizations.

The proof assumes global closure

The proof assumes a declared scope with large-scale coherent witnesses. If the witness family is not standing-bearing or not large-scale coherent, the target changes. If it is, then the coherence burden must be certified.

This conflicts with standard cosmology

It does not. It classifies standard cosmology. If inflationary dynamics supplies the early transport/closure certificate, AASC recognizes its role. The restriction is that the dynamics cannot be used as a primitive origin story or hidden selector.

This cannot predict inflationary observables

Correct. The theorem is structural. n_s , r , A_s , reheating temperature, non-Gaussianities, and e-fold counts are model-specific empirical targets requiring physical dynamics and observational records.

Additional application-facing objections

Objection: the theorem is too abstract to explain inflation

The theorem is abstract because the inflationary role is realized by different possible mechanisms. Scalar-field slow-roll, Starobinsky-type curvature dynamics, bouncing models, varying causal structures, and quantum-gravity mechanisms do not share the same microphysical machinery. They share a burden: late coherent witnesses require a standing-preserving early-to-late certificate when local continuation is insufficient. A theorem at the level of that shared burden must be structural. The abstraction is not a weakness; it is the correct level for a role theorem.

Objection: standard inflation already explains the horizon problem

Standard inflation may indeed supply a physical mechanism in a declared model. AASC does not deny this. It classifies what the mechanism is doing: it supplies a horizon-coherence transport certificate. The AASC theorem explains why some such certificate is structurally required under the target conditions. Standard inflation can then be evaluated as one candidate realization.

Objection: the first metric-comparison regime sounds like a beginning in time

It is not. L_D is a licensing class, not a first instant. It marks the first standing status in which metric comparison is admissible in the declared scope. Temporal language becomes legitimate only after the temporal role is licensed as ordered retained consequence. The phrase “early” is shorthand for dependency position in the interior role order and for its later representation in metric cosmology.

Objection: onset-side closure stress is stipulated

The closure-stress order is defined by successor-neighborhood inclusion. Higher closure stress means fewer admissible standing-preserving successors. The theorem that L_D is onset-side closure-stress maximal is explicitly relative to $\mathcal{O}_D(L_D)$; it does not claim global maximality across terminal or singular regimes. Within that onset-side class, the result follows from the absence of prior metric slack, no proto-metric patchwork, and the requirement of a metric circuit. One may challenge the usefulness of the diagnostic, but the result is not a stipulation once the diagnostic and comparison class are fixed.

Objection: local trajectory insufficiency is just the horizon problem assumed

The theorem is conditional. It does not assert local trajectory insufficiency for every model. It says that if later coherent witnesses are admitted and local trajectory continuation is insufficient, then a certificate is forced. A non-inflationary model may attempt to deny insufficiency by showing that local continuation does suffice. If successful, it avoids the certificate burden by satisfying the local route. If not, it must supply an alternative certificate.

Objection: an alternative model with no inflation refutes the theorem

Only if it also avoids the structural role. A model with no accelerated expansion may still supply ETC_D through another mechanism. That would not refute the theorem; it would realize the structural inflationary role without standard inflation. A model that changes the scope must declare the richer theorem object. A model that assumes coherence without a certificate is hidden load.

Objection: the paper has no empirical predictions

The paper has structural predictions: primitive inflation fails; a late coherent witness target requires an early transport/closure certificate under local insufficiency; scalar fields are optional realization tensors, not structural necessities; reheating must be a role transition. It does not predict numerical observables. That is by design. Numerical predictions belong to physical models that occupy the structural slot.

Objection: the theorem makes inflation unfalsifiable

No. The structural theorem is not the empirical inflation model. Specific realizations remain falsifiable by their observables, parameter constraints, reheating consequences, non-Gaussianities, tensor modes, and consistency relations. The theorem says that if one realization fails, the structural burden remains unless the target conditions are abandoned or another certificate is supplied.

Objection: the theorem privileges global closure

The theorem privileges standing-bearing coherence because that is the target. If a model denies large-scale coherent witnesses, it changes the target. If it accepts them, it must explain their coherence. The certificate may be local, quasi-local, global, nonlocal, or richer-scope depending on the model, but the explanation must preserve standing across the witness family.

Objection: the result is philosophical rather than mathematical

AASC mathematics is not PDE existence theory or Lagrangian model construction. It is fixed-domain constraint exhaustion. The mathematical content is in the definitions, successor-set order, closure theorems, disjunction exhaustions, and counterexample burden. The proof shows that all same-scope routes either supply the certificate or collapse to known normal forms.

Part XIII - Applications, Comparators, and Proof Register

43. Physical realization interfaces

Scalar-field inflation as one realization

The most familiar realization of the inflationary role is a scalar-field model. In ordinary cosmological language one introduces a field ϕ , a potential $V(\phi)$, a background metric, slow-roll conditions, perturbation modes, and a reheating channel. AASC does not deny this model form. It classifies every component by role.

The scalar field is not admitted as a primitive origin source. It is a field-carrier tensor in a declared metric-field-vacuum projection. Its potential is not a free explanatory oracle. It is a model-specific tensor or anchor that controls the field response and the metric-facing source ledger. Slow-roll conditions are not a necessity theorem in AASC; they are a dynamical realization of transport/closure. Reheating is not an unlicensed conversion of pre-metric energy into matter; it is a transition certificate from a field/vacuum carrier regime into particle, gauge, mass, and record-bearing roles.

The key structural requirement is that the scalar model supply the certificate $ETC_D(L_D, W)$. To do that it must show, within its declared scope, how early carrier content is transported to later coherent witnesses without hidden synchronization, repair, or boundary selection. It must also state its anchors: the metric projection, initial or boundary conditions, regularity envelope, field content, gauge treatment, perturbation variable choice, and observational readback. These declarations are not peripheral. They are the conditions under which the model becomes a standing-bearing realization rather than a formal curve fit.

Scope	Status of scalar field ϕ
Metric-only comparison scope	Richer-scope tensor import.
Declared inflationary model scope	Admitted tensor candidate.
Declared scope satisfying ETC_D	Realization of the structural inflationary role.
Undeclared scalar/potential	Hidden load.
Data-fit use without the certificate	Empirical fit without structural closure.

Proposition (scalar-field role status). A scalar-field inflation model is admissible as a realization of the structural inflationary role only if its scalar, potential, background, perturbation variables, reheating channel, and witness readback are declared as anchors or tensors inside the model scope.

Argument. The scalar and its potential do work. If they are not declared, they are hidden tensor load. The background metric and time parameter do work. If they are used before metric

comparison is licensed, they are metric backflow. The reheating channel does work. If it is absent, the model lacks a transition into the particle/gauge/mass roles needed for later records. The perturbation variables and observational readback do work. If they are not controlled, the later witnesses cannot be traced to early carrier content. Therefore the model is admissible only with those role declarations.

This proposition explains why AASC is compatible with scalar-field inflation while refusing to treat it as foundational. A successful scalar model is not the ground of admissibility. It is a declared metric-field-vacuum mechanism that occupies the transport/closure role.

Alternatives to scalar-field inflation

The theorem is intentionally neutral among physical realizations. A bounce model may attempt to supply the early certificate through a pre-contraction or matching condition. An ekpyrotic model may attempt to obtain smoothing and perturbation structure through a different dynamical regime. A varying-speed model may alter the transport envelope. An emergent-spacetime or quantum-gravity model may attempt to replace the early metric picture with a richer-scope structure that projects to metric comparison only later.

AASC does not reject these alternatives. It asks the same questions of each:

1. What is the fixed scope?
2. What late witnesses are being explained?
3. Why is independent local trajectory continuation insufficient?
4. What structure supplies the early transport/closure certificate?
5. Is that structure same-scope, or does it declare a richer theorem object?
6. How are metric comparison, bounded transport, carrier mediation, and record lineage preserved?
7. Which numerical targets are empirical and model-specific?

This turns the inflation debate into a role-comparison problem. Competing cosmological mechanisms are not compared only by whether they reproduce a background expansion history. They are compared by how they certify standing-bearing witness coherence from the earliest licensed comparison regime to later records.

Classification (alternative realization status). Non-standard cosmological models are admissible alternatives to scalar-field inflation only if they either supply $ETC_D(L_D, W)$ in the same scope or explicitly declare a richer scope whose projection preserves the required witness lineage.

Effective globality and why the arrow can look universal

The structural inflationary role is also connected to the arrow-of-time paper. Effective arrows can be plural and need not globally align. Yet in our observed regime they often appear to align: thermodynamic, record, causal, cosmological, and psychological arrows point in a common direction. The early transport/closure certificate supplies one reason such alignment can become practically strong.

If early coherent witness lineage is established across a large comparison domain, later records inherit

a common orientation. Thermodynamic coarse-grainings, structure formation, radiation cooling, and observer memory can then reinforce the same standing order. The alignment is not primitive. It is a physical convergence of effective arrows over a shared record and transport architecture.

AASC therefore distinguishes two claims. The first is structural: an early certificate is required under the late-witness coherence burden. The second is physical: in a particular universe, multiple mechanisms may reinforce that certificate and produce an approximately global arrow. The first is the theorem of this paper. The second is a downstream cosmological program.

Parameter status and numerical nonclaims

Inflationary cosmology is rich in numerical targets: the spectral index n_s , tensor-to-scalar ratio r , scalar amplitude A_s , running, non-Gaussianity, reheating temperature, and the number of e-folds. AASC does not derive these values in this paper. It classifies their status.

A numerical value belongs to the structural theorem only if changing it would change the theorem object. A value belongs to a physical model if it changes a tensor or anchor inside a declared scope. A value belongs to the empirical parameter program if it is fitted, constrained, or inferred from records. Most inflationary numbers are in the last two categories. They are not structural necessities.

Proposition (inflationary parameter discipline). The structural inflationary role theorem does not determine the numerical parameters of a physical inflationary realization. Those values require a declared model, empirical records, calibration ledgers, and exactness or uncertainty certificates.

Argument. The theorem proves the need for a transport/closure certificate under stated conditions. A numerical potential shape, e-fold count, or perturbation amplitude specifies a particular realization of that certificate. Such specification is additional tensor or parameter content. It cannot be read off from the structural theorem without hidden load.

44. Proof architecture and continuation program

Proof spine overview

The proof spine has seven steps.

1. **Boundary exclusion.** Inflation cannot be primitive because it presupposes metric, field, vacuum, energy, transport, and witness roles.
2. **First metric closure.** Licensed metric comparison entails a standing-preserving metric circuit.
3. **No patchwork onset.** Proto-metric local islands require repair, hidden selectors, or incoherent standing.
4. **Successor stress.** Closure stress is defined as thinness of admissible successor neighborhoods.
5. **Onset-side closure stress.** First metric closure has no prior metric slack and is closure-stress maximal relative to the onset-side comparison class.
6. **Witness coherence burden.** Later large-scale coherent witnesses require a standing-preserving lineage.

7. **Certificate necessity.** If local trajectory continuation is insufficient, a same-scope early transport/closure certificate is forced or the explanation collapses.

This sequence is mathematical in the AASC sense. It fixes a target, specifies admissible transformations, exhausts alternatives, and characterizes the surviving structure up to admissible equivalence. It is not constructive model-building.

The fixed-domain comparison class

A common way to misunderstand the theorem is to change the target mid-proof. The target is not every possible cosmological model. It is a fixed admissible realized physical interior with first licensed metric comparison and later standing-bearing large-scale coherent witnesses. A model that denies any of these conditions is not a counterexample; it changes the target.

The same-scope comparison class preserves:

- the AASC kernel K_D ;
- the declared local domains and comparison class;
- the successor relation under fixed constraints;
- the later witness family W ;
- the redescription discipline for records and comparison;
- the no-repair and no-backflow rules.

A proposed explanation that changes one of these may be legitimate, but it is a richer-scope theory, not a same-scope rival.

Endpoint and interior-side interpretation

The beginning/end symmetry is useful only when kept precise. At the end side, exhaustion is $Succ_C(x) = \emptyset$. At the beginning side, the first metric-comparison regime is not the first non-empty successor set in an absolute sense. It is the first standing class in which metric comparison is licensed and closed. Its successor set is non-empty because metric closure exists, but thin on the onset side because there is no prior metric slack and no admissible patchwork repair.

Thus the beginning is not a generative event. It is the first interior-side metric closure. The end is not catastrophe. It is non-continuation. Inflationary structure belongs strictly inside the first case: it is a metric-interior role after comparison is licensed, not a pre-interior cause.

Formal normal forms for failure

Every failed explanation of the inflationary burden falls into one of the following normal forms.

Normal form	Inflation expression	Reason for failure
Skin	coordinate expansion with no standing-bearing transport	represents but does not explain coherence
Hidden load	scalar, boundary, or synchronization rule not declared	does load-bearing work without role status

Normal form	Inflation expression	Reason for failure
Anchor motion	new initial condition or comparison envelope	changes theorem object
Tensor import	extra field, modified gravity, nonlocal dynamics	legitimate only as richer scope
Quotient motion	changes what counts as same perturbation or record	changes witness identity
Continuation motion	changes lawful evolution or transport envelope	changes dynamics/scope
Repair	later fixes incoherent early patches	violates no-repair
Metric backflow	expansion or time before metric licensing	uses downstream representative upstream
Parameter laundering	empirical fit treated as structural proof	confuses evidence with theorem
Standing failure	no certificate for witness coherence	target not met

Mechanism-specific continuation program

The present theorem opens several mechanism-specific research projects. A scalar-field continuation would declare the scalar, potential, gauge, perturbation variables, reheating channel, and empirical witness map. A quantum-gravity continuation would declare the pre-metric or nonmetric richer scope and prove that its projection supplies metric comparison, bounded transport, and records without backflow. A horizon-continuation project would focus on witness-access certificates. A statistical-cosmology continuation would focus on typicality, Past Hypothesis status, and measure anchors.

Each project must satisfy the same protocol. It must identify the late witnesses, explain why local continuation is insufficient, state the certificate, classify anchors/tensors/skin, and declare empirical targets.

45. Detailed literature and standard-cosmology positioning

Inflationary model-building literature

The standard inflationary literature begins with a physical model class. Classic treatments by Guth, Linde, Albrecht-Steinhardt, Starobinsky, Mukhanov-Chibisov, and later effective-field-theory and observational programs ask which potentials, slow-roll conditions, reheating channels, and perturbation spectra fit observed constraints and solve standard cosmological problems. AASC begins one level upstream. It asks why a mechanism with the inflationary role is structurally required if late coherent witnesses are to be explained under insufficient local continuation.

This upstream posture does not compete with model-building. It disciplines it. The scalar field, potential, and background become role-bearing structures rather than primitive origins. A successful

model is one way of realizing the certificate. A failed model may still leave the structural burden intact.

Horizon and flatness problems

The horizon problem is the cleanest bridge between standard cosmology and AASC. Standardly, the problem says that widely separated regions of the CMB appear correlated despite insufficient causal contact in a non-inflationary FLRW history. AASC translates this as witness-lineage insufficiency: later coherent records lack a local-continuation certificate in the candidate non-inflationary metric representative.

The flatness problem similarly becomes a metric-comparison stability burden. To say that curvature remains near a special value is to use a metric representative, comparison class, and dynamical envelope. Inflationary dynamics may stabilize this in a model. AASC says the model must declare how that stability is standing-preserving rather than an assumed representative.

Perturbations and record lineage

Perturbation seeding is where the structural theorem meets observational cosmology. A standard inflationary model treats quantum fluctuations as sources of classical perturbations. AASC asks how that chain becomes a standing-bearing witness lineage. The answer requires a field/vacuum carrier, bounded transport, gauge discipline, decoherence or record-stabilization mechanism, and observational readback. The structural theorem does not compute the spectrum; it states the ledger required for a spectrum to function as evidence.

Past Hypothesis and initial-condition discipline

Inflation is sometimes presented as reducing sensitivity to initial conditions. AASC permits that as a model claim but rejects any primitive initial-condition oracle. A boundary condition can be an anchor. It cannot ground the admissibility conditions that let it be a boundary condition. If inflation is used to explain away special initial data, it must not replace them with an undeclared special scalar state, vacuum selection, or measure choice.

Relation to alternatives to inflation

Alternatives to inflation are often motivated by worries about initial conditions, measure problems, eternal inflation, or high-energy completion. AASC is neutral on their empirical merits. It gives a common burden: if the alternative accepts late coherent witnesses and local continuation insufficiency, it must supply ETC_D . If it denies the insufficiency, it must show how local continuation suffices. If it denies the witness target, it has changed the target.

This makes comparison sharper. The question is not merely which model has a better story. The question is which model supplies a standing-preserving certificate with the least hidden load and clearest empirical witness map.

46. Concrete illustrations

Illustration I: the horizon-coherence burden

The horizon problem is the clearest physical illustration of the theorem. In a non-inflationary metric cosmology, distant regions of the observed microwave background can be represented as lacking ordinary causal contact in the relevant past light-cone structure. The familiar formulation asks how such regions acquired nearly the same temperature. AASC reformulates the question without assuming that ordinary causal contact is the only relevant structure.

Let W_{CMB} be a later record family consisting of standing-bearing microwave background profiles over a large angular domain. Its coherence is not merely a visual pattern. It is a record-stabilized, calibrated, redescription-stable witness family. To use it in a cosmological explanation, the model must identify what counts as the same profile under gauge and observational redescription, how the profile was transported to later records, and which early structures are being inferred.

If independent local trajectory continuation in the candidate non-inflationary metric representative cannot certify the coherence of W_{CMB} , then a same-scope explanation must add a relation that coordinates the relevant representative domains. A standard inflationary expansion supplies one such coordination mechanism by placing the relevant regions inside a common pre-expansion domain in a metric-field-vacuum model. AASC classifies this not as magic expansion from nothing, but as a transport/closure certificate: the later coherent witnesses inherit a common standing lineage from an early metric-comparison regime whose carrier content was coordinated before the local-trajectory insufficiency arises in the non-inflationary representation.

The important point is not that the certificate must look like textbook slow-roll. The important point is that the burden is unavoidable if the target is retained. A model may provide a different mechanism, but it must still explain how the witness family is jointly standing-bearing without hidden synchronization or repair.

Proposition (horizon-coherence application). If a later CMB-scale witness family has standing-bearing coherence and independent local trajectory continuation in the candidate non-inflationary metric representative is insufficient, then any same-scope admissible cosmological account must supply a horizon-coherence certificate, which is a special case of $ETC_D(L_D, W)$.

Argument. The coherence of the CMB-scale record must be more than a display. It must be stable under redescription and available for downstream inference. If local trajectory continuation cannot supply a common lineage, then the common lineage must be supplied by another declared structure. If the structure is standing-preserving and early-to-late, it is an ETC certificate. If it is undeclared, it is hidden synchronization. If it is introduced after incoherence, it is repair. If it changes the scope, it is not a same-scope explanation.

Illustration II: flatness as metric-comparison stability

The standard flatness problem concerns why the observed universe is close to spatial flatness, or why the curvature parameter appears near the value that allows a long-lived large-scale universe. In AASC terms, flatness is not primarily a numerical surprise. It is a metric-comparison stability claim.

A curvature parameter is meaningful only inside a metric projection. It requires a choice of comparison class, a background or effective geometry, a ledger separating coordinate skin from tensor content, and a record path by which the parameter is inferred. To say that curvature remains small across cosmological continuation is to say that a particular metric-comparison feature remains stable across standing-preserving redescription.

Inflationary expansion can make spatial curvature dynamically negligible in many models. AASC does not deny this. It classifies the model as supplying a metric-comparison stability certificate. If a model simply assumes near-flatness as an initial condition, then flatness is a declared anchor. If it uses inflation to reduce sensitivity to curvature, the scalar/vacuum mechanism is a tensor realizing the certificate. If it treats flatness as a primitive fact without anchor status, it is hidden load.

Proposition (flatness certificate). A flatness explanation is admissible only when it declares whether near-flatness is a boundary anchor, a tensor outcome of a transport/closure mechanism, or a downstream empirical fit. Treating flatness as primitive metric authority is backflow.

Argument. Flatness uses metric comparison. Metric comparison is licensed downstream of local-domain and standing structure. A statement about flatness cannot be the source of those conditions. It can be a boundary anchor, a model outcome, or an empirical target. These are distinct claim classes. Failure to classify them turns flatness into a hidden selector.

Illustration III: perturbation spectra and record lineage

Inflationary cosmology is not only a smoothing story. It is also a perturbation story. A standard account takes early quantum fluctuations, evolves them through an inflationary background, freezes or transports them into curvature perturbations, and reads them later through CMB anisotropies or large-scale structure. AASC treats this as a multi-ledger witness-lineage problem.

The early side requires a field/vacuum carrier. The middle requires bounded transport and a gauge discipline that prevents presentation choices from becoming false physical differences. The late side requires record fixation, calibration, and a comparison ledger for spectra. The chain is standing-bearing only if all these pieces are declared and connected without repair.

This is why perturbation spectra are not automatic proof of a primitive inflationary event. They may be evidence for a physical realization of ETC_D , but the standing-bearing content of that evidence depends on the declared model and record ledgers. A different mechanism might produce similar spectra; such a mechanism would be evaluated by whether it supplies the same lineage without hidden load.

Proposition (perturbation-lineage status). A perturbation spectrum functions as evidence for an early transport/closure mechanism only when the model declares the early carrier, transport rule, gauge quotient, record readback, and empirical calibration that connect early structure to later witnesses.

Argument. Without the early carrier, there is no standing-bearing source. Without transport, there is no lineage. Without gauge quotienting, presentation can masquerade as tensor content. Without record readback and calibration, the late spectrum is not an admissible witness. Therefore all ledgers are required.

Illustration IV: reheating as role transition

Reheating is a particularly useful stress test because it connects the early metric-field-vacuum picture to the mature particle/gauge/mass world. A simplistic description says that inflation ends and its energy becomes particles. AASC requires more. The transition must preserve standing while moving from one role region of I_D to another.

The early inflationary structural role primarily concerns comparison, transport, field/vacuum carrier structure, and witness coherence. Later ordinary cosmology requires particles as localized persistence classes, gauge-compatible presentation equivalence, mass/inertia deformation response, thermal records, and eventually complex structures capable of measurement. The reheating role is the bridge between these ledgers.

If the transition is not declared, the model imports particle-bearing structure without explaining how the field/vacuum regime connects to it. If the transition violates gauge or source closure, it changes the theory object. If it erases early records, it performs repair. If it supplies a lawful route into particle/gauge/mass roles, it is a legitimate transition certificate.

Proposition (reheating bridge status). Reheating is admissible as a transition from an early transport/closure certificate to the mature physical interior only when it preserves field-carrier standing, supplies particle/gauge/mass role entry, and maintains record lineage.

Argument. The mature interior cannot be reached by naming particles. Particles, gauge, and mass are roles with deletion tests. A transition that claims to produce the later regime must instantiate those roles or declare the mechanism by which they enter. Therefore reheating is a role bridge, not merely an energy-conversion phrase.

Illustration V: initial conditions as anchors rather than causes

Inflation is often praised for reducing dependence on initial conditions, and criticized for requiring special initial conditions of its own. AASC clarifies the dispute by separating boundary anchors from primitive causes. A boundary condition can be physically important, but it cannot ground the admissibility conditions that make it a boundary condition.

If a model states that a scalar field began in a region of field space with certain properties, that is a boundary or initial data anchor. If it claims that inflation makes a wide basin of initial conditions lead to similar later witnesses, that is a robustness claim inside a declared model. If it claims that no initial condition discipline is needed because inflation itself explains the beginning, it backflows.

The appropriate AASC question is not whether initial conditions are good or bad. It is whether their role is declared. A declared anchor can be evaluated. An undeclared selector cannot.

Proposition (initial-condition role discipline). Initial conditions in inflationary cosmology are admissible as declared anchors or empirical/model assumptions. They are inadmissible as primitive generators of standing, metricity, or directionality.

Part XIV - Conclusion

Inflation is not primitive. It is not an admissibility-generating event, a pre-metric cause, or a privileged initial-condition oracle. Inflationary language presupposes the very metric, temporal, field, vacuum, transport, and record structures it might be tempted to explain.

Yet AASC does have real explanatory power. First metric comparison cannot be licensed as incoherent patchwork. It requires standing-preserving closure. Because no prior metric slack is available at first closure, the successor neighborhood is maximally thin relative to the onset-side comparison class; this is onset-side closure stress. Under that stress, ordinary local trajectory continuation cannot carry all global coherence burdens. If later large-scale coherent witnesses have standing, their lineage must be certified. That certificate is the early transport/closure role.

In metric-field-vacuum projection language, the early transport/closure certificate is the structural inflationary role. Standard inflationary models may realize it. Alternative cosmological mechanisms may realize it differently. Empirical data may support or constrain particular realizations. The structural theorem remains: any same-scope account of late coherent witnesses must either supply early metric-interior transport closure, declare a boundary anchor, declare a richer-scope tensor mechanism, or collapse into hidden selection, repair, scope change, metric backflow, or standing failure.

In compressed form:

$$FirstMetricClosure \Rightarrow OnsetSideClosureStress \Rightarrow ThinOnsetSuccessorSet$$

and

$$LateCoherentWitnesses \wedge LineageBurden \wedge LocTrajInsufficient \Rightarrow ETC_D \Rightarrow InflRole_D$$

inside a metric-field-vacuum projection.

AASC therefore neither proves nor denies empirical inflation. It proves the structural slot that inflation-like physics is asked to occupy.

Appendices

Appendix A - Source dependency index

Source cluster	Claim type	Function	Boundary
META/KERNEL	imported constraint layer	admissibility, standing, reference, irreversibility	no physical ontology

Source cluster	Claim type	Function	Boundary
AMetric/ATS	interface discipline	no primitive metric, coordinate, time, probability, energy	no field equation
Beginning of the Interior	early boundary module	first metric licensing, no proto-metric patchwork, onset-side closure stress	older terminology modernized
End of the Interior	successor-set module	successor set and exhaustion notion	no temporal end narrative
Alpha-Omega capstone	closure framing	beginning/end as structural bounds	not a full cosmology
RPI	realized-physics role architecture	local domains, comparison, bounded transport, fields, vacuum, time, causality	not an inflation paper
Arrow of Time	no-backflow and cosmological arrow discipline	standing monotonicity, effective cosmological arrows, initial-condition firewall	not an inflation proof
Horizon and cosmological transport arcs	support family	witness access, reconstruction, transport closure	no full empirical model
ZPE/vacuum arc	support family	vacuum-offset standing discipline	no dark-energy solution

Appendix B - Symbol index

Symbol	Meaning
D	fixed admissible realized physical interior
C	fixed admissibility/continuation constraint set
K_D	AASC kernel: admissibility, standing, reference, irreversibility
I_D	realized physical interior role package
L_D	first licensed metric-comparison regime
$Succ_C(x)$	standing-preserving admissible successor set of x under C
$ClStress_C(x)$	closure-stress diagnostic of x as successor-neighborhood thinness

Symbol	Meaning
$\mathcal{O}_D(L_D)$	onset-side comparison class for first metric closure
$R_D, \tau_D, \chi_D, \mathcal{S}_D$	components of the early transport/closure certificate
W	later large-scale coherent witness family
$LineageBurden(W, L_D)$	explanatory requirement that the coherence of W be accounted for from L_D
$ETC_D(L_D, W)$	early transport/closure certificate tuple $(R_D, \tau_D, \chi_D, \mathcal{S}_D)$
R_D	early standing-bearing relation over L_D
τ_D	bounded transport relation from early content to later witnesses
χ_D	closure certificate for witness coherence
$\mathcal{S}_D$	scope ledger declaring anchors, tensors, and skin
$\mathcal{O}_D(L_D)$	onset-side first-closure comparison class
$InflRole_D(W)$	metric-field-vacuum projection of the early certificate
Cmp_D	quotient-stable comparison role
$BoundTrans_D$	bounded perturbation transport role
Φ_D	field-carrier role
Ω_D	vacuum baseline role

Appendix C - Standard inflation interface and role taxonomy

Standard object	AASC role	Status
$\ddot{a} > 0$	accelerated-expansion representative	metric projection, not primitive
$a(t)$	scale-factor representative	admissible only after metric/time projection
$H = \dot{a}/a$	metric/time derivative representative	downstream of comparison and temporal roles
$(aH)^{-1}$	horizon/reachability representative	possible display of transport burden
ϕ	field-carrier tensor	optional richer-scope realization
$V(\phi)$	model-specific tensor/anchor	not derived structurally
ρ_ϕ, p_ϕ	metric-facing source ledger	declared physical model content
$N = \int H dt$	e-fold count	model-specific numerical target
perturbations	record/witness lineage	empirical/model-specific

Standard object	AASC role	Status
reheating	transition certificate	required if model uses field/vacuum-to-particle bridge
multiverse/eternal inflation	scope expansion	not same-scope theorem

Appendix D - Claim-class ledger

Claim class	Examples	Control rule
Paper-level theorem	first metric closure, onset-side closure-stress maximality, ETC necessity	prove from imported roles
Controlled correspondence Physical model	inflation, slow-roll, reheating scalar field, bounce, ekpyrotic	classify as possible realization declare anchors/tensors/witnesses
Empirical target	n_s, r, A_s, N	no claim without model and data
Nonclaim	origin of admissibility, why universe exists	outside scope

Appendix E - Extension protocol

Any extension of this paper should answer:

1. What is the fixed scope?
2. What are the late coherent witnesses?
3. Why is local trajectory continuation insufficient?
4. What is the proposed transport/closure certificate?
5. Is the certificate same-scope or richer-scope?
6. Which anchors, tensors, and skin are declared?
7. How are record lineage and gauge/metric representatives controlled?
8. Which empirical targets are model-specific rather than structural?
9. What collapse route is avoided?
10. What would falsify the proposed realization?

Appendix F - Detailed source scan summary

The local corpus scan used for this manuscript produced the following high-level result. Literal occurrences of “inflation” in the matrix were not reliable indicators of cosmological inflation; most referred to exactness inflation, count inflation, scalar-sector inflation, hierarchy inflation, or interface-authority inflation. The relevant support instead came from role-neighboring source clusters.

Cluster	Manuscript use
AMetric boundary	blocks pre-admissible metric/time/energy use
Metricity as licensed description	supports first metric-comparison regime
Metric role from continuation comparison	makes metricity secondary to local domains and comparison
Continuation loci/local domains	supplies pre-metric local organization
Bounded perturbation transport	central to early-to-late witness lineage
Fields as persistence carriers	supplies field projection language
Vacuum as minimal concrete persistence	supplies non-null no-particle baseline
Gauge-vacuum compatibility	controls vacuum/transition claims
Horizon instability	supplies witness-access and reconstruction discipline
Initial conditions are not physical inputs	blocks primitive boundary selection
Cosmological transport closure	supports early-to-late inference discipline
Arrow of Time	supplies no-backflow and cosmological effective-arrow status
Realized Physical Interior	supplies the role ladder used throughout
Beginning/End/Alpha-Omega	supply boundary concepts and successor-set closure-stress language

Appendix G - Counterexample burden

A successful counterexample must preserve the fixed target while avoiding all collapse routes. It must exhibit a fixed admissible realized physical interior D with first metric-comparison regime L_D and later large-scale coherent witnesses W , show that independent local trajectory continuation is insufficient, and nevertheless avoid an early transport/closure certificate. It must also avoid hidden selector import, repair, scope change, metric backflow, energy backflow, field/vacuum backflow, and standing failure.

A model that simply denies late coherent witnesses is not a counterexample. A model that supplies a different mechanism is not a counterexample if that mechanism occupies the certificate role. A model that changes the early domain is a richer-scope alternative. A model that fits data without the lineage ledger is parameter evidence, not structural rebuttal.

Appendix H - Glossary

Term	Meaning in this manuscript
admissibility	licensed status of construction or description in a fixed scope
standing	reusable status of admitted content under lawful downstream use
redescription	lawful presentation change preserving standing-bearing content

Term	Meaning in this manuscript
backflow	use of a downstream representative to authorize its own licensing conditions
first metric-comparison regime	first standing class in which quotient-stable comparison is licensed
metric circuit	standing-preserving redescription loop for metric surrogates
successor set	admissible standing-preserving continuations under fixed constraints
closure stress	successor-neighborhood thinness diagnostic; not physical stress-energy unless a metric projection is declared
transport/closure certificate	structure certifying early-to-late coherent witness lineage
structural inflationary role	metric-field-vacuum projection of the transport/closure certificate
richer scope	declared enlargement of anchors, tensors, or admissibility conditions
skin	presentation surplus with no independent standing-bearing load

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